

HERO scorecard — value ± BCa 95% CI · cell colour = per-metric min-max (green = best), * = Δ-vs-frozen CI excludes 0
8 encoders × 10 metrics · bold + WINNER row = single best per metric · WINS partitions the 9 scorable metrics (Σ = 9) · 10 K bootstrap

	action_top1 ↑ better	motion_cos ↑ better	taxonomy_f1 ↑ better	future_mse ↓ better	rollout ↓ better	causal ↓ better	tdist ↓ better	teacher_free ↓ better	maskratio ↓ better	order ± signed	WINS ↑ better
V-JEPA 2.1 frozen	44.384±2.274	0.009±0.001	0.793±0.016	0.557±0.001	0.009±0.000	0.583±0.001	0.007±0.000	0.046±0.000	0.072±0.001	0.018±0.001	1/9
pretrain_2X encoder	53.151±2.274*	0.169±0.004*	0.786±0.017	0.546±0.001*	0.009±0.000*	0.571±0.001*	0.008±0.000*	0.044±0.001*	0.071±0.001*	0.021±0.001*	1/9
pretrain encoder	51.671±2.274*	0.159±0.004*	0.791±0.017	0.540±0.001*	0.008±0.000*	0.565±0.001*	0.007±0.000	0.043±0.001*	0.068±0.001*	0.019±0.001*	3/9
pretrain head	42.849±2.280	0.013±0.001*	0.793±0.017	0.558±0.001*	0.009±0.000	0.583±0.001	0.007±0.000	0.046±0.000	0.072±0.001	0.018±0.001	0/9
surgery 3stage_DI encoder	50.301±2.274*	0.163±0.004*	0.793±0.017	0.515±0.001*	0.009±0.000*	0.547±0.002*	0.008±0.000*	0.048±0.001*	0.059±0.001*	0.019±0.002*	2/9
surgery 3stage_DI head	51.233±2.301*	0.167±0.005*	0.789±0.017	0.540±0.001*	0.008±0.000*	0.565±0.001*	0.007±0.000	0.043±0.001*	0.068±0.001*	0.019±0.001*	0/9
surgery noDI encoder	51.781±2.274*	0.168±0.005*	0.790±0.017	0.515±0.001*	0.009±0.000*	0.545±0.002*	0.008±0.000*	0.047±0.001*	0.062±0.001*	0.021±0.001*	1/9
surgery noDI head	51.726±2.301*	0.179±0.005*	0.788±0.017	0.540±0.001*	0.008±0.000*	0.565±0.001*	0.007±0.000	0.043±0.001*	0.068±0.001*	0.019±0.001*	1/9
WINNER / metric	pre-2X	sNoDI-hd	frozen	s3DI-enc	pre-enc	sNoDI-enc	pre-enc	pre-enc	s3DI-enc	—	Σ 9/9